Fiscal Year:	FY 2021	**Task Last Updated:**	FY 05/02/2021
PI Name:	Blakely, Eleanor A Ph.D.		
Project Title:	Simulation of GCR-Induced Harderian Gland and Lung Tumorigenesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	eablakely@lbl.gov	**Fax:**	FY 510-486-4475
PI Organization Type:	UNIVERSITY	**Phone:**	510-486-6595
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	One Cyclotron Road		
PI Address 2:	MS977		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	02/04/2016	**End Date:**	03/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2022 and grant number is now 80JSC02IT0017, per PI (Ed., 6/7/21) NOTE: End date changed to 2/4/2022 per PI and Space Radiation element (Ed., 3/17/21) NOTE: End date is 2/4/2021 per PI (Ed., 11/7/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mao, Jian-Hua Ph.D. (Lawrence Berkeley National Laboratory) Chang, Polly Ph.D. (SRI International)		
Grant/Contract No.:	80JSC02IT0017 ; NNJ16HP22I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Increased cancer risk remains one of the primary concerns for travel into deep space and may preclude manned missions to Mars due to the large uncertainties that currently exist in estimating cancer risk from the spectrum of radiation types found in space with available human epidemiological data on radiation-induced cancer. Existing data on human risk of cancer from exposures to X- and gamma-rays must be scaled to the many types and fluxes of radiations found in space using radiation quality factors and dose-rate modification factors. Linearity of dose-response is often assumed since the shape of the dose-response at doses below 100 mSv is unknown. However, preliminary analysis of our recently completed experiments with silicon and titanium ion suggest non-linear structure in the low-dose region for both Harderian and lung tumorigenesis. The goal of this proposal is to reduce uncertainties in the estimation of particle radiation carcinogenesis from galactic cosmic radiations (GCR). Our approach is to use the existing data on Harderian gland tumorigenesis from investigations of individual high-atomic number, and high-energy (HZE) ion beams to predict outcome of mixed ion chronic GCR exposures, and then measure Harderian gland and lung dose-dependent tumor data for such exposures. We will compare some of the measured cancer prevalence outcomes of a GCR-simulated exposure with the theoretical predictions based on experiments with individual ion beams. Our main hypothesis is that HZE carcinogenic effects are LET- (linear energy transfer) and dose-dependent and not additive at low particle doses due to the contribution of non-targeted effects. Completion of the proposed GCR studies using a mouse strain that has already been well characterized with a low spontaneous tumor background will allow a significant test of modeling approaches used to estimate carcinogenesis risk from a simulated GCR beamline.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Completion of our research on predicting and measuring HZE tumorigenic effects to test the hypothesis that HZE carcinogenic effects and LET- and dose-dependent are not additive at low particle doses due to the contribution of non-targeted effects will allow a significant test of modeling approaches used to estimate carcinogenic risk from single-ion and mixed radiation fields. This work will also benefit estimation of risk of tumorigenesis from mixed radiation fields on Earth.
Task Progress:	Recent experimental milestones A. Three-HZE ion Study: A total (including sham-treated controls) of 419 female CB6F1 mice were exposed to a 3- heavy ion mixed beam during the spring of 2019, delivered in rapid sequence to a total dose of 30 or 60 cGy. After irradiation, animals were shipped from Brookhaven National Laboratory to the vivarium at Lawrence Berkeley National Laboratory (LBNL) for 16-month long-term housing and maintenance until their scheduled necropsy. Animals that were moribund or were observed to bear tumors were removed early based on LBNL's IACUC (Institutional Animal Care & Use Committee) protocol sacrifice criteria. Tumor tissues were collected and fixed for potential down-stream analysis. For this reason, the actual number of animals received at SRI International for necropsy were less than the starting total number of animals. Harderian Gland (HG), lung tissues, and other tissues with gross abnormalities were collected, fixed, and processed to slides for histological evaluations. To ensure that microscopic disease is discovered in the HG gland, each gland that is embedded in the blocks was step sectioned to exhaustion. Whenever possible, portions of the HG and lung tissues were snap frozen and stored at -20oC for future analysis. Other major organs such as kidneys, liver, lung, spleen, and brain were frozen and fixed for downstream processing or for sharing with other NASA investigators. During necropsy, bone marrow lymphocytes from 10 animals in each group were extracted, cultured, and fixed on slides. Slides were shipped to Dr. Megumi Hada's laboratory for fluorescent in situ hybridization (FISH) staining with chromosomes 1, 2, and 4 whole chromosome probes to determine long-term aberration prevalence in these animals. Rolls of colon tissues from 10 animals in each group were harvested, fixed with formalin, and processed to slides. The slides were shipped to Dr. Jerry Shay's laboratory for colon polyp assessment. No colon tumors were found either in control or irradiated animals. Ovarian tissues were collected, fixed, and shipped to Dr. Ulrike Lederer's laboratory for analysis. Sample analysis work is in progress and results will be reported in the next progress report. In silico prediction with incremental effect synergy theory of 3-HZE ions was modeled, based on results for 2-ion beams (Huang et al., LSSR 25:107-118, 2020, and Huang & Sachs, LSSR 26:173-174, 2020). Calculations and estimates were made assuming non-targeted effect (NTE) + targeted effect (TE) action, but assuming TE only action gave similar results:prevalences of ~23% for 30 cGy, and ~36% for 60 cGy total doses. Synergy theory modeling of the actual tumor prevalences using TE modeling indicated that there were higher values than predicted- Level 1* synergy at 30 cGy, while TE + NTE modeling showed Level 2* synergy at 30 cGy. At scheduled necropsy, bone marrow-derived lymphocytes from 10 animals in each group were cultured, fixed, and shipped to our collaborator, Dr. Megumi Hada for FISH chromosome aberration (CA)-scoring. Results showed that the percent of apparently simple and total exchanges and the number of color junctions were significantly elevated 16 months after radiation for both dose regimes of the 3 HZE ion exposures, although the % complex exchanges were not significantly different from the controls. These results speak to persistent long-term space radiation effects in the bone marrow hematopoietic compartment that resulted in the accumulation of damaged cells with simple exchanges, but not those with complex exchanges. B. Six-ION GCRsim Study: We irradiated animals to a single total dose of 100 cGy of a 6-ion GCRsim in the Spring of 2018. Here we report outcome data on tumor histology, prevalence, synergy modeling, and chromosome aberrations (CA) in bone marrow lymphocytes at 16-months post exposure. We found significantly increased HG tumor prevalence in GCRsim irradiated animals when compared to the controls ($p=0.026$). Although benign adenomas are the dominant tumor-type, the HG in 1 control animal was found to be a lymphoma while 1 animal in the irradiated group had a HG carcinoma. We found changes in lung tumor incidence and pathology in 6-ion-GCRsim-irradiated animals but the change was not significant when compared to the controls ($p=0.41$ obtained by Fisher's exact test). Although benign adenomas are also the dominant tumor-type in the lungs, there were more carcinomas in the GCRsim irradiated animals than in the controls. In addition to the HG and lung tumors, ovarian tumor incidences were also elevated in the irradiated animals. No spontaneous tumors were found in the control cohort. A variety of tumors, including luteoma (with luteal-like cells that

	are bigger than a corpus luteum and compresses the surrounding tissue), mixed tumors, lipoma, cystadenomas, tubulostromal adenomas, and granular cell tumors that are either benign or malignant were found in the GCRsim irradiated animals. Other sporadic tumors in the GCR-irradiated animals include renal tubule adenomas, pituitary adenoma. Professor Rainer Sachs and his University of California Berkeley team completed theoretical modeling of the data obtained in the 6-ion beam GCRsim Harderian Gland tumor prevalence study, to test for synergy or antagonism. A modern synergy theory, Incremental Effect Additivity (IEA), was used, for the reasons detailed in the paper by Huang et al. and its web supplement (Huang et al., LSSR 25:107-118, 2020). Tumor prevalence results of the 6-ion beam exposure with a total mixture dose of 100 cGy was measured to yield HG tumor prevalence of ~23.5%. The curves and confidence intervals (CI) assumed HZE NTE action dominates at very low doses, so that there is a smooth, but very steep, highly concave increase of prevalence from 0 dose to very small doses. Swift light ion effects were modeled by a single Incremental Effect Additivity (IEA) curve prediction for the prevalence if no synergy or antagonism is present among the components of the mixed beam, and a 95% CI, which was calculated appropriately, taking adjustable parameter correlations into account. Our results suggest synergy is more likely than antagonism, since the actual tumor prevalence lies above the predicted IEA line. However, because the error bar of the data point overlaps the CI band, the data indicate—no significant synergy among the 6 beams. The results from the analysis of murine CA in bone marrow of mice harvested at 16 months post irradiation with the 6-ion GCRsim beam mixture by Dr. Hada and her team showed that there was a significant increase in % total ($p=0.007$ obtained by Mann-Whitney test) and simple exchanges ($p=0.007$ obtained by Mann-Whitney test), but no significant increase in complex exchanges ($p=0.74$ obtained by Mann-Whitney test) in the 6-ion GCRsim exposure when compared to the unirradiated controls.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Abstracts for Journals and Proceedings	Chang PY, Bakke J, Grover A, Rusek A, Snyder D, Rhone JR, Parra M, Bjornstad KA, Mao JH, Hada M, Sachs RK, Blakely EA. "Murine Harderian Gland (HG) Tumorigenesis After 3 HZE-Mixed Field Exposure." Presented at the 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Online Proceedings. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021
Abstracts for Journals and Proceedings	Awodele O, Mishra B, Chang P, Blakely E, Luderer U. "Ovarian Effects of Space Radiation in Comparison to Gamma Radiation." Presented at the 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Online Proceedings of the 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021
Abstracts for Journals and Proceedings	Chang PY, Bakke J, Grover A, Rusek A, Synder D, Mao JH, Parra M, Sachs RK, Blakely EA. "Murine Harderian Gland (HG) Tumorigenesis: 6-Beam GCR Simulation Exposure." Presented at the 66th Annual Meeting of Radiation Research Society, Virtual Meeting, October 18-21, 2020. Online Proceedings and short presentation. 66th Annual Meeting of Radiation Research Society, Virtual Meeting, October 18-21, 2020. , Oct-2020
Abstracts for Journals and Proceedings	Blakely EA. "Radiation Effect on the Cell Cycle and its Checkpoints." Presented at the 66th Annual Meeting of Radiation Research Society, Virtual Meeting, October 18-21, 2020. Online Proceedings of the 66th Annual Meeting of Radiation Research Society, Virtual Meeting, October 18-21, 2020. , Oct-2020
Abstracts for Journals and Proceedings	Hada M, Rhone JR, Plante I, Slaba T, Patel Z, Chang PY, Blakely EA, Mao JH, Bakke J, Doppalapudi R, Grover A, Wang S, Shi Y, Sachs R, Saganti PB. "Chromosome Aberrations Induced by GCR-Simulated Mixed-Beam Exposure in Human and Mouse Cells--Computational Model Prediction and Biological Validation." Presented at the Committee on Space Research (COSPAR) 2021-Hybrid, 43rd Scientific Assembly, Sydney, Australia, January 28-February 4, 2021. Online Proceedings of the Committee on Space Research (COSPAR) 2021-Hybrid, 43rd Scientific Assembly, Sydney, Australia, January 28-February 4, 2021. , Feb-2021
Articles in Peer-reviewed Journals	Garikipati VNS, Arakelyan A, Blakely EA, Chang PY, Truongcao MM, Cimini M, Malaredy V, Bajpai A, Addya S, Bisserier M, Brojakowska A, Eskandari A, Khigatian MK, Hadri L, Fish KM, Kishore R, Goukassian DA. "Long-term effects of very low dose particle radiation on gene expression in the heart: Degenerative disease risks." Cells. 2021 Feb 13;10(2):387. https://doi.org/10.3390/cells10020387 ; PMID: 33668521; PMCID: PMC7917872 , Feb-2021
Articles in Peer-reviewed Journals	Huang EG, Sachs RK. "Commentary on "Simulating galactic cosmic ray effects: Synergy modeling of murine tumor prevalence after exposure to two one-ion beams in rapid sequence." [Supplements the article: Huang EG et al. Simulating galactic cosmic ray effects: Synergy modeling of murine tumor prevalence after exposure to two one-ion beams in rapid sequence. Life Sci Space Res. 2020 May;25:107-18.https://pubmed.ncbi.nlm.nih.gov/32414484]" Life Sci Space Res (Amst). 2020 Aug;26:173-4. https://doi.org/10.1016/j.lssr.2020.03.007 ; PMID: 32718683 , Aug-2020
Articles in Peer-reviewed Journals	Blakely EA. "The 20th Gray Lecture 2019: Health and heavy ions." Br J Radiol. 2020 Nov 1;93(1115):20200172. https://doi.org/10.1259/bjr.20200172 ; PMID: 33021811 , Nov-2020
Awards	Blakely EA. "The 20th Gray Medal of the ICRU (International Commission on. Radiation Units & Measurements), presented at the 16th International Congress of Radiation Research, Manchester, England, August 15-29, 2019." Aug-2019